

NANTUCKET, MA, USA

WMO: 725060

Lat: 41.253N

Lon: 70.061W

Elev: 15

StdP: 101.15

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 14756

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.1	-7.7	-18.8	0.7	-8.5	-16.7	0.9	-6.6	16.4	0.7	14.9	0.8	6.9	310	0.628

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.4	26.4	21.5	25.1	21.2	24.0	20.6	23.1	24.9	22.4	24.0	21.8	23.2	5.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	17.2	24.1	22.0	16.7	23.5	21.3	16.0	22.7	68.5	25.0	65.9	23.9	63.7	23.3	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.8	12.0	10.9	-13.1	29.8	3.0	1.7	-15.2	31.0	-17.0	31.9	-18.6	32.9	-20.8	34.1		
(5)			-14.2	24.5	2.8	0.9	-16.2	25.1	-17.8	25.6	-19.3	26.1	-21.4	26.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.5	0.6	0.9	3.2	7.6	12.2	16.7	20.6	20.7	17.8	13.0	8.1	3.7
	DBStd	7.86	4.78	3.84	3.56	2.69	2.72	2.68	2.10	2.13	2.68	3.41	3.80	4.25
	HDD10.0	1134	291	254	211	79	9	0	0	0	0	12	79	199
	HDD18.3	3054	548	488	468	321	190	63	5	4	41	167	306	454
	CDD10.0	1313	0	0	1	8	78	200	329	331	236	105	22	3
	CDD18.3	194	0	0	0	0	1	13	76	78	26	2	0	0
	CDH23.3	402	0	0	0	2	3	29	174	169	23	2	0	0
	CDH26.7	37	0	0	0	0	1	4	19	12	1	0	0	0
Wind	WSAvg	5.5	6.2	6.2	6.4	5.9	5.3	4.8	4.4	4.2	4.8	5.7	5.9	6.0
Precipitation	PrecAvg	1017	96	84	82	83	86	64	61	72	88	85	95	116
	PrecMax	1453	187	142	161	184	264	174	167	171	241	189	189	247
	PrecMin	643	30	19	17	34	15	6	2	8	2	22	30	42
	PrecStd	247	36	36	41	39	64	45	48	44	59	44	40	51
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.4	10.9	13.1	17.5	22.5	26.5	28.8	28.2	25.3	22.2	17.4	14.7
		MCWB	10.8	9.7	9.9	11.4	16.8	20.1	22.7	22.2	20.8	18.8	15.3	13.1
	2%	DB	9.9	9.0	11.0	14.2	19.6	23.7	26.8	26.3	23.9	20.2	16.1	12.8
		MCWB	9.0	7.7	8.6	10.4	15.1	19.0	21.6	22.0	20.4	18.0	14.6	11.9
	5%	DB	8.7	7.8	9.5	12.9	17.7	22.2	25.2	25.1	22.8	19.0	15.0	11.6
		MCWB	7.6	6.6	7.7	9.9	14.1	18.4	21.2	21.7	20.0	17.3	13.3	10.6
	10%	DB	7.5	6.8	8.3	11.8	16.4	20.8	23.9	24.0	22.1	17.9	13.8	10.2
		MCWB	6.3	5.3	6.7	9.3	13.3	17.9	20.6	21.0	19.5	16.2	12.3	9.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.9	10.0	11.2	13.7	18.0	21.9	24.0	24.5	22.8	20.0	16.4	13.6
		MCDB	11.1	10.5	12.0	15.5	21.2	25.1	27.0	26.5	24.2	21.0	16.8	14.2
	2%	WB	9.3	8.1	9.3	12.0	16.5	20.3	22.9	23.4	21.8	18.7	15.0	12.1
		MCDB	9.6	8.6	10.2	13.1	18.7	22.4	25.2	25.0	22.9	19.5	15.8	12.5
	5%	WB	7.9	7.0	8.1	10.9	15.2	19.3	22.2	22.7	21.1	17.8	13.8	10.8
		MCDB	8.4	7.7	8.9	11.9	16.9	21.1	24.0	24.1	22.0	18.5	14.7	11.4
	10%	WB	6.5	5.6	7.0	9.9	14.0	18.4	21.5	22.1	20.3	16.8	12.6	9.1
		MCDB	7.3	6.6	8.0	11.1	15.6	19.9	23.1	23.3	21.3	17.8	13.5	10.0
Mean Daily Temperature Range	5% DB	MDBR	6.3	6.5	6.3	6.5	6.7	6.3	6.3	6.4	6.6	6.4	6.4	6.5
		MCDBR	6.9	7.4	7.2	8.1	8.8	8.5	7.7	6.9	6.9	6.6	6.7	7.1
		MCWBR	7.2	6.9	5.9	5.3	5.2	4.9	4.1	3.9	4.6	5.3	6.2	7.6
	5% WB	MCDBR	6.9	7.2	6.7	6.8	7.8	6.9	6.3	5.8	5.8	5.9	6.2	7.3
		MCWBR	7.4	7.3	6.3	5.6	5.3	4.6	3.9	3.8	4.6	5.5	6.4	8.1
Clear-Sky Solar Irradiance	taub	0.321	0.327	0.345	0.384	0.418	0.465	0.487	0.447	0.377	0.357	0.328	0.316	
	taud	2.494	2.481	2.464	2.368	2.313	2.220	2.167	2.281	2.443	2.501	2.544	2.539	
	Ebn at Noon	834	886	905	888	861	818	794	818	859	839	820	811	
	Edh at Noon	75	88	100	118	127	140	146	127	101	85	71	67	
All-Sky Solar Radiation	RadAvg	1.68	2.61	3.71	4.84	5.48	5.89	6.03	5.45	4.45	3.03	1.97	1.45	
	RadStd	0.14	0.25	0.30	0.52	0.54	0.52	0.33	0.32	0.34	0.25	0.16	0.14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+104	+51

Nomenclature: See separate page